Spivak Chapter 2 Problems

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1 Problem 1

Prove the following formulas by induction.

1. $1^2 + ... + n^2 = \frac{n(n+1)(2n+1)}{6}$.

   For $k = 1$, the formula holds. Assume the above holds for all k. Now, considering $k+1$, we have

$$1^2 + ... + k^2 + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2,$$
$$\frac{k(k+1)(2k+1) + 6(k+1)^2}{6},$$
$$\frac{(k+1)[k(2k+1) + 6(k+1)]}{6},$$
$$\frac{(k+1)(k+2)(2(k+1)+1)}{6}.$$

2. $1^3 + ... + n^3 = (1 + ... + n)^2$.

   For $k = 1$, the formula holds. Assume the above holds for all k. Now, considering $k+1$, we have

$$1^3 + ... + k^3 + (k+1)^3 = (1 + ... + k)^2 + (k+1)^3,$$
$$(1 + ... + k)^2 + (k+1)(k+1)^2,$$
$$\frac{(k(k+1))^2}{4} + (k+1)(k+1)^2,$$
$$\frac{(k(k+1))^2 + 4(k+1)(k+1)^2}{4},$$
$$\frac{(k+1)^2[(k^2 + 4(k+1)]}{4},$$
$$\frac{(k+1)^2(k+2)^2}{4},$$
$$\left(\frac{(k+1)(k+2)}{2}\right)^2.$$

2 Problem 2

Find a formula for

1. $\sum_{i=1}^{n}(2i-1) = 1+3+5+...+(2n-1)$.

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2},$$
$$\sum_{i=1}^{n} 2i = n(n+1),$$
$$\sum_{i=1}^{n}(2i-1) = n(n+1) - n = n^2.$$

3 Problem 5

1. Prove by induction on n that

$$1+r+r^2+...+r^n = \frac{1-r^{n+1}}{1-r}$$

   if $r \neq 1$.

   Assume $r \neq 1$. For $k=1$, the formula holds. Assume the above holds for all k. Now, considering $k+1$, we have

$$1+r+r^2+...+r^k+r^{k+1} = \frac{1-r^{k+1}}{1-r} + r^{k+1},$$
$$\frac{1-r^{k+1}+(1-r)(r^{k+1})}{1-r},$$
$$\frac{1-r^{k+2}}{1-r}.$$

2. Derive this result by setting $S = 1+r+...+r^n$, multiplying this equation by r, and solving the two equations for S.

$$S = 1+r+r^2+...+r^n, Sr = r+r^2+...+r^{n+1},$$
$$S - Sr = 1 - r^{n+1},$$
$$S(1-r) = 1-r^{n+1},$$
$$S = \frac{1-r^{n+1}}{1-r}.$$

4 Problem 8

Prove that every natural number is either even or odd.

An even number is a number of the form $2n$, an odd number $2n+1$. For 1, $1 = 2 \cdot 0 + 1$, so 1 is odd. Assume k is either even or odd. For odd k, we have $k+1 = 2n+1+1 = 2(n+1)$, an even number. For even k, we have $k+1 = 2n+1$, an odd number. Thus, every natural number is either even or odd.

5 Problem 12

1. If a is rational and b is irrational, is $a+b$ necessarily irrational? What if a and b are both irrational?

   Assume $a+b$ is rational; then $(a+b)-a$ would be rational, because the rationals are closed under subtraction. Thus, we have a contradiction. However, the sum of two irrationals is not necessarily irrational: consider the expression $\sqrt{2} + (2 - \sqrt{2}) = 2$.

2. If a is rational and b is irrational, is ab necessarily irrational?

   Yes, for $a \neq 0$. The rationals are closed under multiplication.

6 Problem 13

1. Prove that $\sqrt{3}, \sqrt{5}$ and $\sqrt{6}$ are irrational.

   Suppose $\sqrt{3}$ is rational. Then there is some $(\frac{p}{q})^2 = 3$, or $p^2 = 3q^2$. Every integer can be expressed either in the form $3n$, $3n+1$ or $3n+2$. Because $3q^2$ is of the form $3n$, p^2 must also be of the form $3n$, as if 3 divides p^2 it must divide p. Thus, $9k^2 = 3q^2$, so $3k^2 = q^2$, which contradicts the assumption that one of p, q is not a multiple of 3. Thus, $\sqrt{3}$ is irrational.